

Backend Development & Configurations

1) Creation of roles

This module involved creating custom roles to control access within the Smart Library Request Workflow application. Two roles were created — librarian and student — under User Administration → Roles. The librarian role is intended for staff who manage book issue/return operations, while the student role is intended for users who raise borrow requests. Both roles were verified to appear correctly in the Roles module list, ensuring they are available for assignment to users and for use in Access Control Lists (ACLs) later in the project.

The screenshot shows the 'Role - New Record' form in ServiceNow. The 'Name' field is filled with 'librarian', the 'Application' is set to 'Global', and the 'Elevated privilege' checkbox is unchecked. The 'Description' field is empty. A 'Submit' button is at the bottom left.

The screenshot shows the 'Role - New Record' form in ServiceNow. The 'Name' field is filled with 'student', the 'Application' is set to 'Global', and the 'Elevated privilege' checkbox is unchecked. The 'Description' field is empty. A 'Submit' button is at the bottom left.

The screenshot shows the 'Roles' list view in ServiceNow. The table has columns for 'Name', 'Description', and 'Elevated privilege'. Two roles are listed: 'u_book_user' and 'u_borrow_request_user', both with 'Elevated privilege' set to 'false'.

Name	Description	Elevated privilege
u_book_user		false
u_borrow_request_user		false

2) Create Tables

Two custom tables were created under System Definition → Tables to store the core application data. The Book table (u_book) stores details of each book available in the library, and the Borrow Request table (u_borrow_request) stores each request raised by a student to borrow a book. Both tables were created with the Global application scope and verified to appear correctly under the Tables module, forming the foundational data structure for the workflow.

The first screenshot shows the 'Table - New Record' form for the 'Book' table. The 'Label' is 'Book' and the 'Name' is 'u_book'. The 'Application' is set to 'Global'. The 'Create module' and 'Create mobile module' checkboxes are checked. The 'Add module to menu' dropdown is set to '-- Create new --' and the 'New menu name' is 'Book'. The 'Remote Table' checkbox is unchecked.

The second screenshot shows the 'Table - New Record' form for the 'Borrow Request' table. The 'Label' is 'Borrow Request' and the 'Name' is 'u_borrow_request'. The 'Application' is set to 'Global'. The 'Create module' and 'Create mobile module' checkboxes are checked. The 'Add module to menu' dropdown is set to '-- Create new --' and the 'New menu name' is 'Borrow Request'. The 'Remote Table' checkbox is unchecked.

The third screenshot shows the 'Tables' list in the ServiceNow interface. It displays two tables: 'Book' and 'Borrow Request'. The 'Book' table has a label of 'Book', a name of 'u_book', and an 'Extends table' of '(empty)'. The 'Borrow Request' table has a label of 'Borrow Request', a name of 'u_borrow_request', and an 'Extends table' of '(empty)'. Both tables are marked as 'Extensible' and were last updated on 2026-07-06 02:44:41 and 2026-07-06 02:45:29 respectively.

Label	Name	Extends table	Extensible	Updated
Book	u_book	(empty)	false	2026-07-06 02:44:41
Borrow Request	u_borrow_request	(empty)	false	2026-07-06 02:45:29

3)Add Fields to Book table

The Book table (u_book) was configured with the following fields to capture complete book information:

- Title (String) – Name of the book
- Author (String) – Author of the book
- ISBN (String) – Unique book identification number
- Status (Choice) – Availability status of the book, with choice values *Available*, *Issued*, *Lost*, and a default value set to *Available*

These fields ensure that every book record carries enough detail for identification and for tracking whether it is currently available to be borrowed. The Status field default value of "Available" ensures newly added books are automatically marked as available for issue.

servicenow All Favorites History Workspaces Admin Dictionary Entry - Status

Search

Dictionary Entry Status

Create Choice List Delete Column Update

Choice List Specification Default Value

Displays a list of suggested values in a **Choice** list. In the Advanced view you can select the **Choice** table and the **Choice** field to take choice values from, plus a **Dependent** field.

Choice Dropdown without -- None -- (must specify a default value)

Create Choice List Delete Column Update

Related Links

Show Table Run Point Scan Advanced view

Access Controls Choices (3) Attributes Labels (1)

Label Search

Actions on selected rows... New

Label	Value	Language	Sequence	Inactive	Updated
Lost	100	en		false	2026-07-06 02:54:58
Issued	100	en		false	2026-07-06 02:54:33
Available	100	en		false	2026-07-06 02:54:16

1 to 3 of 3

4)Add Fields to Borrow Request table

The Borrow Request table (u_borrow_request) was configured with the following fields to capture each borrowing transaction:

- Requested By (Reference to User [sys_user]) – Identifies the student/user raising the request
- Book (Reference to Book [u_book]) – Identifies which book is being requested
- Request Date (Date/Time) – Records when the request was made
- Status (Choice) – Tracks the state of the request, with choice values *Requested*, *Approved*, *Rejected*, *Returned*, and a default value set to *Requested*

These fields allow the system to link a student, a book, and a timestamp together for every borrow transaction, and to track its lifecycle from initial request through to return.

servicenow						
All Favorites History Workspaces Admin						
Choice - New Record						
Search						
Table Borrow Request						
New menu name						
Remote Table						
Columns Controls Application Access						
Table Columns for text Search						
Dictionary Entries						
Column label	Type	Reference	Max length	Default value	Display	
Created	Date/Time	(empty)		40	false	
Sys ID	Sys ID (GUID)	(empty)		32	false	
Updates	Integer	(empty)		40	false	
Request Date	Date/Time	(empty)		100	false	
Requested By	Reference	User		32	false	
Updated by	String	(empty)		40	false	
Updated	Date/Time	(empty)		40	false	
Book	Reference	Book		32	false	
Created by	String	(empty)		40	false	
Status	Choice	(empty)		100 Requested	false	

5)Configure Reference Qualifier (Prevent Requesting Already-Issued Books)

To prevent students from requesting books that are already issued or lost, a Reference Qualifier was configured on the Book field of the Borrow Request table. A Simple reference qualifier was applied with the condition Status is Available. This ensures that the Book lookup field on the Borrow Request form only displays books whose status is "Available," automatically filtering out books that are currently issued or marked as lost. This configuration enforces data integrity and prevents duplicate/invalid borrow requests at the source, without requiring additional validation scripts.

servicenow		All Favorites History Workspaces Admin		Dictionary Entry - Book		Search			
Dictionary Entry Book View: Advanced						Delete Column		Update	
Alters the behavior of a field or functionality that depends on the field. More Info									
Attributes									
Reference Specification Choice List Specification Dependent Field Calculated Value Default Value									
The Reference field specifies what table this field displays values from.									
* Reference Book									
Use reference qualifier Simple									
Reference qual condition Add Filter Condition Add OR Clause									
Status is Available AND OR									
Reference Specification - Additional Customization									
Reference key									
Reference cascade rule -- None --									
Reference floats									
Dynamic creation									
Delete Column Update									